

Algebra II with Trigonometry

Chapter 6.2

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4506907_alg-2trig-h-chp-6-2-note-docx.pdf](#)

Here is a comprehensive math guide based on the provided document.

1. Short Summary of the Document

This document covers Section 6.2, "Trigonometric Functions of Real Numbers." It introduces the six trigonometric functions based on the coordinates of a terminal point $P(x, y)$ on the unit circle. The text explains the domain of these functions, how to determine their positive or negative signs based on the four quadrants, and how to evaluate them using reference numbers. Finally, it provides fundamental identities (reciprocal, Pythagorean, and even-odd properties) and a set of blank examples for students to practice finding terminal points, signs, quadrants, composite functions, and even/odd symmetry.

2. Core Definitions, Theorems, and Formulas

Core Definitions:

- **Trigonometric Functions (Circular Functions):** If t is a real number and $P(x, y)$ is the terminal point on the unit circle determined by t :
- $\sin t = y$
- $\cos t = x$
- $\tan t = \frac{y}{x}$ (where $x \neq 0$)
- $\csc t = \frac{1}{y}$ (where $y \neq 0$)
- $\sec t = \frac{1}{x}$ (where $x \neq 0$)
- $\cot t = \frac{x}{y}$ (where $y \neq 0$)

- **Domains:**
- $\sin, \cos$: All real numbers.
- $\tan, \sec$: All real numbers *except* $\frac{\pi}{2} + n\pi$ (where n is an integer).
- $\cot, \csc$: All real numbers *except* $n\pi$ (where n is an integer).

Theorems & Properties:

- **Signs by Quadrant:**
- **Quadrant I:** All positive.
- **Quadrant II:** $\sin, \csc$ are positive (others negative).
- **Quadrant III:** $\tan, \cot$ are positive (others negative).
- **Quadrant IV:** $\cos, \sec$ are positive (others negative).
- **Even-Odd Properties:**
- **Even Functions:** $\cos(-t) = \cos t$ and $\sec(-t) = \sec t$.
- **Odd Functions:** $\sin(-t) = -\sin t$, $\csc(-t) = -\csc t$,
 $\tan(-t) = -\tan t$, $\cot(-t) = -\cot t$.

Core Formulas (Fundamental Identities):

- **Reciprocal Identities:**
- $\csc t = \frac{1}{\sin t}$
- $\sec t = \frac{1}{\cos t}$
- $\cot t = \frac{1}{\tan t}$
- $\tan t = \frac{\sin t}{\cos t}$
- $\cot t = \frac{\cos t}{\sin t}$
- **Pythagorean Identities:**
- $\sin^2 t + \cos^2 t = 1$
- $\tan^2 t + 1 = \sec^2 t$
- $1 + \cot^2 t = \csc^2 t$

3. Worked Study Guide

Here is a step-by-step breakdown of the most important problem types found in the document's examples.

Concept A: Finding Trig Values Given a Point on the Unit Circle (Based on Ex 2)

- **The Rule:** If you are given a terminal point (x, y) , remember that $x = \cos t$ and $y = \sin t$.
- *Worked Example:* Given the point $(\frac{3}{5}, \frac{-4}{5})$, find $\sin t$, $\cos t$, and $\tan t$.
- *Step 1:* Identify x and y . Here, $x = \frac{3}{5}$ and $y = \frac{-4}{5}$.
- *Step 2:* $\cos t = x$, so $\cos t = \frac{3}{5}$.
- *Step 3:* $\sin t = y$, so $\sin t = \frac{-4}{5}$.
- *Step 4:* $\tan t = \frac{y}{x}$. Divide the y -value by the x -value: $\frac{-4/5}{3/5} = \frac{-4}{3}$. So, $\tan t = -\frac{4}{3}$.

Concept B: Using Clues to Determine the Quadrant (Based on Ex 4)

- **The Rule:** Use the "Signs of the Trigonometric Functions" table.
- *Worked Example:* Find the quadrant where $\csc t > 0$ and $\sec t < 0$.
- *Step 1:* Analyze the first clue. $\csc t > 0$ means cosecant is positive. Cosecant is positive in Quadrants I and II.
- *Step 2:* Analyze the second clue. $\sec t < 0$ means secant is negative. Secant is negative in Quadrants II and III.
- *Step 3:* Find the overlap. Both conditions are only true in **Quadrant II**.

Concept C: Using Pythagorean Identities (Based on Ex 5)

- **The Rule:** You can write one function in terms of another using $\sin^2 t + \cos^2 t = 1$ or its variations. Pay close attention to the quadrant to determine if the final square root should be positive or negative.
- *Worked Example:* Write $\csc t$ in terms of $\cot t$ if the terminal point is in Quadrant II.
- *Step 1:* Choose the identity connecting them: $1 + \cot^2 t = \csc^2 t$.
- *Step 2:* Take the square root of both sides to isolate $\csc t$:

$$\csc t = \pm \sqrt{1 + \cot^2 t}.$$
- *Step 3:* Determine the sign. In Quadrant II, $\csc t$ (which pairs with sine) is positive. Therefore, the answer is $\csc t = \sqrt{1 + \cot^2 t}$.

Concept D: Testing for Even/Odd Functions (Based on Ex 8)

- **The Rule:** Substitute $(-x)$ into the function. If $f(-x) = f(x)$, it is Even. If $f(-x) = -f(x)$, it is Odd.

- *Worked Example:* Is $f(x) = x^2 \cos(2x)$ even, odd, or neither?
 - *Step 1:* Plug in $(-x)$: $f(-x) = (-x)^2 \cos(2(-x))$.
 - *Step 2:* Simplify using algebra and Even-Odd properties. $(-x)^2 = x^2$. For cosine, $\cos(-2x) = \cos(2x)$ because cosine is an even function.
 - *Step 3:* The result is $x^2 \cos(2x)$, which is exactly our original $f(x)$.
Because $f(-x) = f(x)$, the function is **Even**.
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4. Five Practice Problems

These problems are taken directly from the unworked examples in the provided PDF.

Problem 1: Find the exact value of $\tan(15\pi)$. (*From Example 1b*) **Problem**

2: The terminal point determined by real number t is $(\frac{-6}{7}, \frac{-\sqrt{13}}{7})$. Find $\sin t$, $\cos t$, and $\tan t$. (*From Example 2b*) **Problem 3:** From the given

information, find the quadrant in which the terminal point determined by t lies: $\cos t > 0$ and $\cot t < 0$. (*From Example 4b*) **Problem 4:** Find functions f and g such that $F = f \circ g$ given $F(x) = \sqrt{1 + \tan x}$. (*From Example 7b*)

Problem 5: Determine whether the function is even, odd, or neither:

$f(x) = x^3 + \cos x$. (*From Example 8b*)

Answers to Practice Problems:

1. 0. (The terminal point for 15π is the same as π , which is $(-1, 0)$).

$\tan t = y/x = 0/-1 = 0$). 2. $\sin t = \frac{-\sqrt{13}}{7}$, $\cos t = \frac{-6}{7}$, $\tan t = \frac{\sqrt{13}}{6}$. 3.

Quadrant IV. ($\cos$ is positive in I, IV. $\cot$ is negative in II, IV. The overlap is IV).

4. $g(x) = 1 + \tan x$ and $f(x) = \sqrt{x}$. (Note: $g(x) = \tan x$ and

$f(x) = \sqrt{1+x}$ is also a valid answer). 5. **Neither.** Plugging in $(-x)$ yields $f(-x) = (-x)^3 + \cos(-x) = -x^3 + \cos x$. This is not exactly $f(x)$ (not even) and is not exactly $-f(x)$ which would be $-x^3 - \cos x$ (not odd).

5. Assumptions and Ambiguities Resolved

- **Assumption regarding Exact Values:** The document outlines the *steps* to evaluate trig functions (Reference Number $\rightarrow$ Sign $\rightarrow$ Value) but does not print a table of standard unit circle values (like $\pi/6, \pi/4, \pi/3$). I

assumed the student has access to a unit circle or has learned these standard coordinates in section 6.1.

- **Resolution of Image text vs. OCR:** In Example 2b, the raw text scan read $-\frac{13}{7}$, but looking closely at the provided image, the numerator is clearly under a square root symbol: $-\frac{\sqrt{13}}{7}$. I used the correct visual data from the image to formulate the study guide and practice problems.
- **Example 9 Ambiguity:** Example 9 asks to find the displacement $y(t) = 4 \cos(3\pi t)$ at specific decimals (0, 0.25, etc.). I did not include this in the practice section as evaluating it requires converting decimals to fractions (e.g., $t = 0.25 \rightarrow 3\pi/4$) and knowing the exact unit circle value for that angle, which relies heavily on outside knowledge not explicitly printed on these 5 pages.